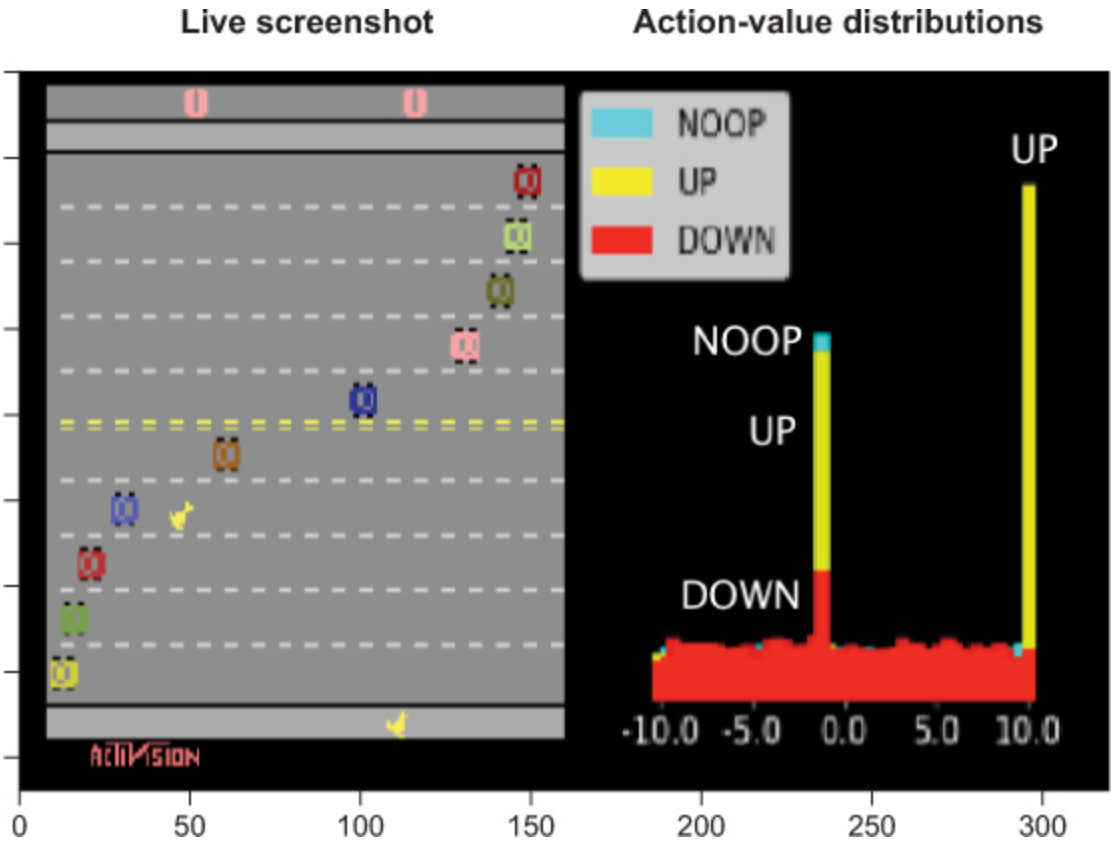
occurred. If you're getting four or more, that indicates a successfully trained agent.

Figure 7.21 shows a mid-training game screenshot alongside the corresponding predicted action-value distributions (again, refer to the GitHub code to see how to do this).

**Figure 7.21. Left: Screenshot of live gameplay in Atari Freeway. Right: The corresponding action-value distributions of each of the each actions overlaid. The spike on the right corresponds to the UP action and the spike on the left corresponds mostly to the NO-OP action. Since the right spike is larger, the agent is more likely to take the UP action, which seems like the right thing to do in this case. It is difficult to see, but the UP action also has a spike on top of the NO-OP spike on the left, so the UP action-value distribution is bimodal, suggesting that taking the UP action might lead to either a –1 reward or a +10 reward, but the +10 reward is more likely since that spike is taller.**




In figure 7.21 you can see that the action-value distribution for the UP action has two modes (peaks): one at –1 and the other at +10. The expectation value of this distribution is much higher than the other actions, so this action will be selected.